

Designed by Remus Teodorescu 2026

LECTURE 7.2

Fundamental PDEs

Remus Teodorescu

With support from Research Assistant Noman Khan - nomank@energy.aau.dk

Aalborg University · ret@et.aau.dk

Deep Learning for Engineering · 2026

CLASSIFY · STEADY · IN TIME

Poisson	elliptic: a source, every wall
Helmholtz	elliptic: a mode and its eigenvalue
heat	parabolic: one initial field
wave	hyperbolic: two initial conditions

Part 2 is built on four equations. Poisson and Helmholtz are steady: a condition on every wall and none in time, and Helmholtz asks for a mode and its eigenvalue together. Heat and wave add time as an input, and one or two initial conditions.

Learning Objectives

SECTIONS, IN ORDER

1. **Classification** — the three types, and the flow equations
2. **Poisson** — the steady reference, and its conditions
3. **Helmholtz** — a mode and an eigenvalue, and u ≡ 0
4. **Heat** — time as an input, and one initial field
5. **Wave** — two initial conditions, and sine activations

READ, AND WHAT EACH ANSWERS

Liu ch. 2	which conditions make it well posed?
Liu ch. 2.4	what does a steady balance need?
Liu ch. 6	how is a mode found, not zero?
Liu ch. 4.1	what does time add to Poisson?
Liu ch. 5	why two initial conditions?

PDE Classification

PDE classification — a second-order linear equation is elliptic, parabolic or hyperbolic according to the sign of its discriminant $B^2 - 4AC$.

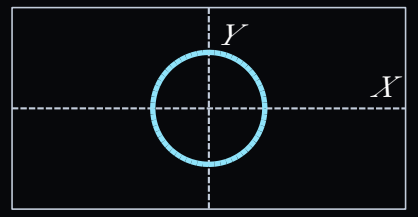
ONE DISCRIMINANT, THREE TYPES

	Δ	example	what must be given
elliptic	< 0	Poisson	the edges
parabolic	= 0	heat	the edges + the start field
hyperbolic	> 0	wave	the edges + start field + start speed

$$A\,u_{xx} + B\,u_{xy} + C\,u_{yy} + D\,u_x + E\,u_y + F\,u = G \qquad \Delta = B^2 - 4AC$$

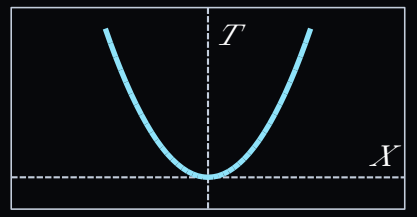
$A \ldots G$ the coefficients, constants or functions of x, y; Δ the discriminant

POISSON · ELLIPTIC



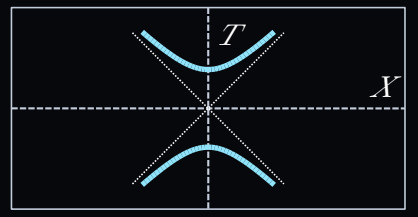
$X^2 + Y^2 = 1$ · *ellipse*

HEAT · PARABOLIC



$T = \alpha X^2$ · *parabola*

WAVE · HYPERBOLIC



$T^2 - c^2 X^2 = 1$ · *hyperbola*

WHAT EACH ONE NEEDS

Poisson — steady: the edges set the inside, so give the edges

heat — changes in time: also give where it starts

wave — swings like a spring: also give its start speed

Helmholtz — elliptic, like Poisson; λ is unknown too, and $u \equiv 0$ always solves it

The rule: one initial condition per time derivative.

Fluid Flow PDEs

Fluid flow PDEs — Navier–Stokes, $\rho(\mathbf{u}_t + \mathbf{u}\cdot\nabla\mathbf{u}) = -\nabla p + \mu\nabla^2\mathbf{u}$ with $\nabla\cdot\mathbf{u} = 0$, moves a fluid through open space; Darcy, $\mathbf{u} = -(K/\mu)\nabla p$, moves it through a porous solid. Neither is one pure type: each is built from the three.

THE TWO FLOW EQUATIONS, CLASSIFIED

equation	type	used for
Navier–Stokes, slow	parabolic, like heat	a channel, a pipe
Navier–Stokes, fast	convection: carried, like a wave	a wake, turbulence
Darcy, steady	elliptic: Poisson for p	a porous electrode
Darcy, in time	parabolic: pressure diffuses	groundwater

TALK · 2 MIN How would you explain, in your own words, why one flow can behave like heat and like a wave? Then, in the card on the right, which term takes over as the Reynolds number grows, and what does the problem then need?

WHICH TERM BRINGS WHICH TYPE

time term — parabolic: a start field is needed

viscous term — elliptic: every wall matters

convective term — carries the flow along, like a wave

continuity — a Poisson equation for the pressure

The Reynolds number, convection over viscosity, decides which type wins: low, the flow smooths like heat; high, it carries like a wave.

Poisson Equation

Poisson equation — the steady balance between diffusion and a distributed source, $u_{xx} + u_{yy} + f = 0$, with no time. It is elliptic: every point feels every boundary at once, so it needs a value or a flux at every boundary point and no initial condition.

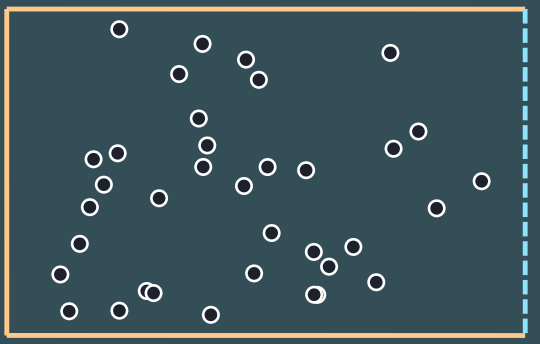
KIND, CONDITIONS, APPLICATIONS

- **kind** — elliptic, no time
- **conditions** — a value or a flux on every wall
- **applications** — conduction, electrostatics

$$u_{xx} + u_{yy} + f(x,y) = 0 \qquad u = \bar{u} \text{ on } \partial\Omega_D, \quad \frac{\partial u}{\partial n} = \bar{q} \text{ on } \partial\Omega_N$$

domain	conditions	exact solution
rectangle	u on every wall	$u = \Sigma A_{mn} \sin(m\pi x/a) \sin(n\pi y/b)$
rectangle	u and flux (mixed)	$u = \Sigma A_{mn} \sin(m\pi x/a) \cos((n - \frac{1}{2})\pi y/b)$
disc	u on the rim, f const	$u = f(R^2 - r^2)/4$
any shape	flux on every wall	$u + C$, C any constant
irregular	any	none: numerical (L8.1)

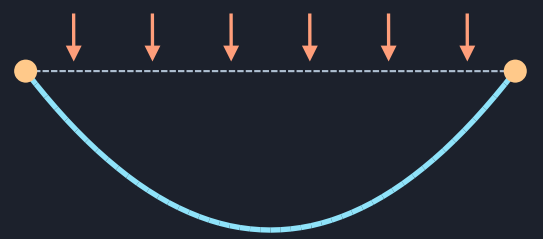
A MIXED PROBLEM



solid: Dirichlet · dashed: Neumann

f the source, $\bar{u}$ the value on $\partial\Omega_D$, $\bar{q}$ the flux on $\partial\Omega_N$

A MEMBRANE UNDER A LOAD



a load f pushes

the edge is held

the field u sags

Helmholtz Equation

Helmholtz equation — Poisson's equation with the source replaced by λu: $\nabla^2 u + \lambda u = 0$. Only certain pairs of a mode u and an eigenvalue λ solve it, so both are unknown together, and a mode is fixed only up to its scale. It arises whenever the wave equation is separated in time.

KIND, CONDITIONS, APPLICATIONS

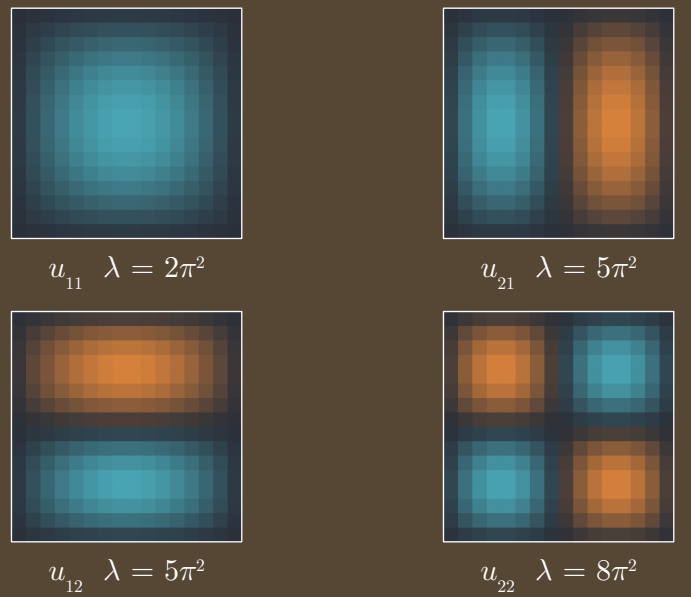
- **kind** — an eigenvalue problem
- **conditions** — the walls, and a fixed norm
- **applications** — vibration and acoustic modes

$$\nabla^2 u + \lambda u = 0, \quad \lambda = k^2 \qquad u_{mn} = 2\sin(m\pi x)\sin(n\pi y), \quad \lambda_{mn} = (m^2 + n^2)\,\pi^2$$

λ the eigenvalue, k the wavenumber, u_{mn} the mode with m and n half waves

domain	conditions	exact solution
rectangle	$u = 0$ on the walls	$u = \sin(m\pi x/a)\,\sin(n\pi y/b), \quad \lambda = \pi^2(m^2/a^2 + n^2/b^2)$
rectangle	no flux on the walls	$u = \cos(m\pi x/a)\,\cos(n\pi y/b), \quad$ same λ
disc	$u = 0$ on the rim	$u = J_m(kr)\,\cos(m\theta), \quad J_m(kR) = 0, \quad \lambda = k^2$
irregular	any	none: numerical

FIRST FOUR MODES OF THE UNIT SQUARE



Each seen from above, like a drum skin.

Trivial Solution

Trivial solution — $u \equiv 0$ solves the Helmholtz equation and its walls for any λ, with a residual of exactly zero, and gradient descent finds it first. The loss must exclude it: a normalisation term fixes the scale and a floor keeps λ away from zero.

A PERFECT SCORE FOR A WRONG ANSWER

$$u \equiv 0, \quad \lambda = 0$$

$u \equiv 0$ the zero field : it satisfies the equation and the walls for any λ

- **the expected outcome** — not a rare failure
- **two safeguards** — a unit norm and a floor on λ

THE SAFEGUARDS IN PYTORCH

```
L_norm = (u.pow(2).mean() - 1).pow(2)          # unit norm: u ≡ 0 fails it
lam = lam0 + nnf.softplus(lam_raw)             # a floor: λ stays above λ₀
loss = L_pde + L_dbc + w_n * L_norm            # zero only for a true mode
```

WHAT SATISFIES WHICH TERM

	residual	walls	norm = 1
$u \equiv 0, \lambda = 0$	yes	yes	no
mode (1,1)	yes	yes	yes
mode (2,1)	yes	yes	yes
2 × mode (1,1)	yes	yes	no

Only the norm column tells the trivial solution from a mode. On the unit square the lowest eigenvalue is $2\pi^2 \approx 19.74$: a λ far from it is a bug.

Orthogonality

Orthogonality — two modes with different eigenvalues are orthogonal: the integral of their product over the domain is zero. Asking each new mode to be orthogonal to those already found, and of unit norm, makes each run find the next mode instead of the lowest again.

ONE MODE AT A TIME

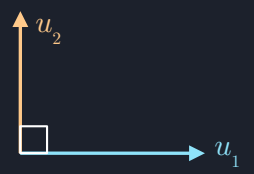
$$\langle u_{mn}, u_{pq} \rangle = \int_\Omega u_{mn}\, u_{pq}\, d\Omega = \delta_{mp}\, \delta_{nq}$$

u_{mn} the mode (m, n); δ is one when its indices agree, zero otherwise

$$\mathcal{L} = \mathcal{L}_{PDE} + w_o\, \mathcal{L}_{orth} + w_n\, \mathcal{L}_{norm}, \quad \lambda \geq \lambda_0$$

w_o, w_n the weights of the two extra terms, λ_0 the floor on λ

ORTHOGONAL, DRAWN



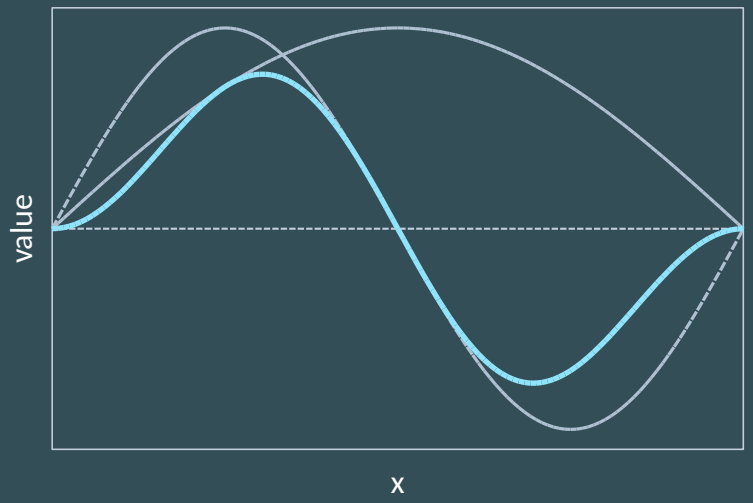


modes are like perpendicular axes

a mode with itself: one

with any other: zero

ORTHOGONALITY, VISUALLY



Grey sin πx, dashed sin 2πx, cyan their product: equal positive and negative area, so its integral is zero.

Heat Equation

Heat equation — the transient version of Poisson's equation, $u_t = \alpha(u_{xx} + u_{yy}) + f$. It is parabolic, first order in time: besides the boundary for all t it needs one initial field, time becomes a third input, and every disturbance spreads and decays.

KIND, CONDITIONS, APPLICATIONS

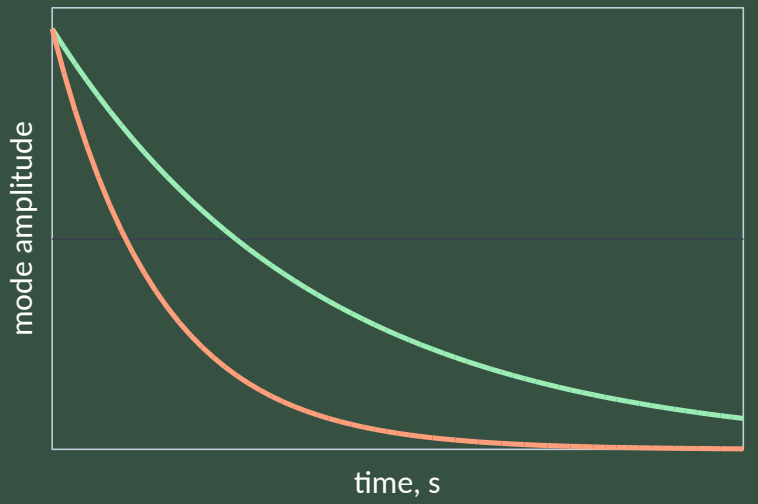
- **kind** — parabolic, first order in time
- **conditions** — the walls, and one initial field
- **applications** — a die after a power pulse

$u_t = \alpha\,(u_{xx} + u_{yy}) + f(x, y, t)$ $\qquad$ $u(x, y, 0) = f_{IC}(x, y)$

α the diffusivity, f the source, f_{IC} the field at $t = 0$

domain	conditions	exact solution
rectangle	$u = 0$ on the walls	$u = \Sigma A_{mn}\, e^{-\alpha\lambda_{mn}t} \sin(m\pi x/a) \sin(n\pi y/b)$
rectangle	no flux on the walls	$u = \Sigma A_{mn}\, e^{-\alpha\lambda_{mn}t} \cos(m\pi x/a) \cos(n\pi y/b)$
infinite line	point pulse at $t = 0$	$u = e^{-x^2/4\alpha t}/\sqrt{4\pi\alpha t}$
irregular	any	none: numerical (L8.2)

THE DIE'S TWO MODES DECAY



Ex_07.2's die: mode (1,1) in green decays at 17.4 /s, mode (2,1) in orange at 43.4 /s, so the field changes shape, not only size.

Wave Equation

Wave equation — $u_{tt} = c^2(u_{xx} + u_{yy}) + f$: a disturbance travels at the finite speed c and keeps its shape instead of decaying. It is hyperbolic, second order in time, so it needs the boundary and two initial conditions. The panel struck in Ex_07.2 rings at 141 Hz.

KIND, CONDITIONS, APPLICATIONS

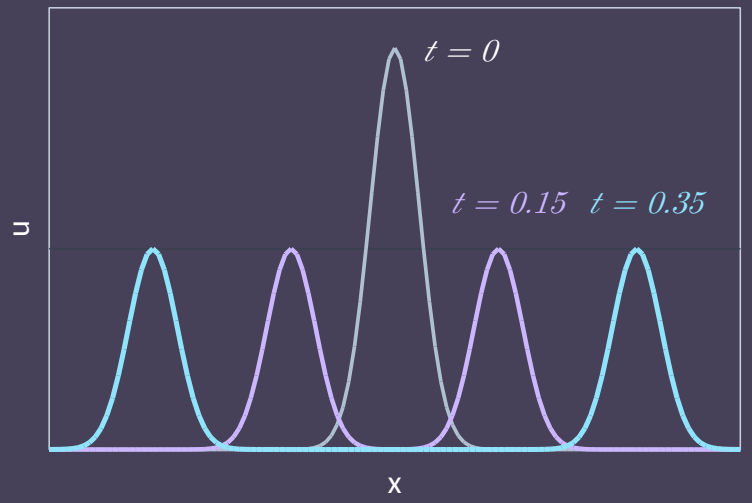
- **kind** — hyperbolic, second order in time
- **conditions** — the walls, a shape and a velocity
- **applications** — a struck panel, acoustics

$$u_{tt} = c^2\,(u_{xx} + u_{yy}) + f(x, y, t) \qquad u(x, y, 0) = u_0(x, y), \quad u_t(x, y, 0) = v_0(x, y)$$

c the wave speed, u_0 the initial shape, v_0 the initial velocity

domain	conditions	exact solution
rectangle	$u = 0$ on the walls	$u = \Sigma A_{mn} \cos(\omega_{mn} t + \varphi_{mn}) \sin(m\pi x/a) \sin(n\pi y/b)$
rectangle	no flux on the walls	$u = \Sigma A_{mn} \cos(\omega_{mn} t + \varphi_{mn}) \cos(m\pi x/a) \cos(n\pi y/b)$
infinite line	a shape and a speed	$u = f(x - ct) + g(x + ct)$
irregular	any	none: numerical

A PULSE SPLITS AND TRAVELS



d'Alembert's solution: the pulse splits into two halves that travel at c and keep their shape. Nothing decays.

Two Initial Conditions

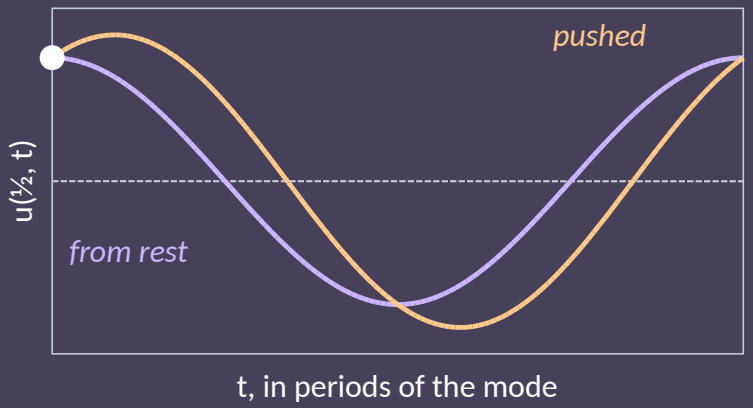
Initial velocity — the second initial condition of an equation that is second order in time: the rate u_t at $t = 0$. The same starting shape released from rest or given a push has two different futures, so without it the problem is under-determined.

THE SAME SHAPE, TWO FUTURES

- **the rule** — as many initial conditions as time derivatives
- **left out** — u ≡ 0 satisfies everything else
- **the loss** — converges, and the panel never moves

TALK · 2 MIN How would you explain why a second-order equation needs two initial conditions, in your own words? Then, in the plot on the right, what would a network trained without the velocity condition produce, and why would its loss look good?

u AT THE CENTRE, AGAINST TIME



THE VELOCITY IN PYTORCH

```
L_iv = (u_t0 - v_0).pow(2).mean()    # velocity at t = 0
loss = L_pde + L_dbc + L_ic + L_iv   # drop L_iv: u ≡ 0
```

Periodic Activations

Periodic activations — a first layer that uses $\sin(\omega_0\, z)$ instead of tanh. A wave is a sum of sinusoids, and the derivative of a sinusoid is again a sinusoid, so the network can hold oscillations that a tanh network smooths into their envelope.

MATCH THE NETWORK TO THE SOLUTION

$$\phi(\mathbf{x}) = \sin(\omega_0\, \mathbf{W}\mathbf{x} + \mathbf{b}), \quad \omega_0 \approx 30$$

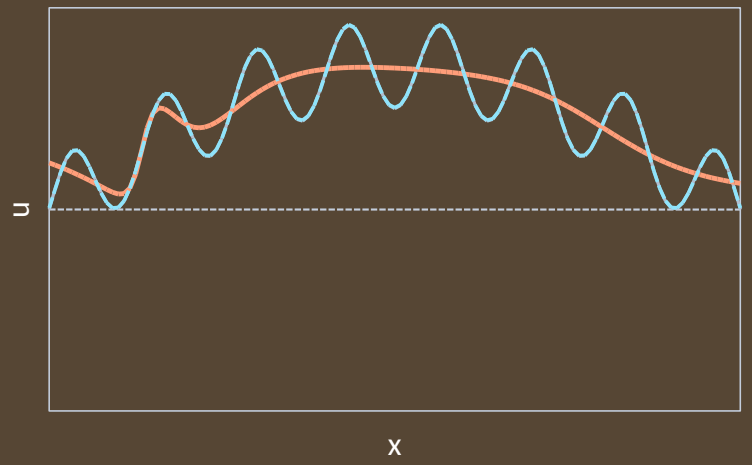
ω_0 the frequency scale of the first layer, **W**, **b** its weights and biases

- **oscillations** — kept, not smoothed away
- **a fair test** — the same size, the same steps

A SINE LAYER IN PYTORCH

```
class Sine(nn.Module):                              # the periodic activation
    def forward(self, z): return torch.sin(30 * z) # ω₀ = 30
net = nn.Sequential(nn.Linear(3, 64), Sine(), nn.Linear(64, 1))  # sine layer
```

MEASURED · TANH AGAINST SINE



tanh 32 x 3	2209 weights, error 25.7 %
sine 32 x 3	2209 weights, error 0.53 %

Orange tanh, dashed cyan sine, grey the target. 3000 Adam steps, lr 1e-3, seed 0.

Advanced Topics

RELATED TOPICS LEFT OUT, AND WHERE THEY MATTER

topic	what it adds	read	where it makes a difference
learned boundary	hard walls without a polynomial	Liu ch. 3.1 – 3.3	a plate with a hole
space–time training	the whole time block at once	Liu ch. 4.2 – 4.3	long transients
causal training	early times fitted first	Liu ch. 4.4	fast transients
many modes	K modes, each orthogonal	Liu ch. 6.5	vibration spectra

WHY THEY ARE LEFT OUT

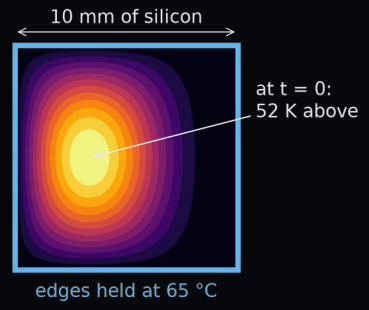
Each returns with a device: the plate with a hole and the transient die in L8, and Burgers' equation and its shock with the flow of L9. None is needed to solve today's four equations.

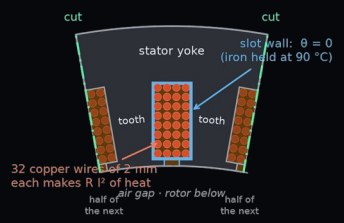
EXERCISE

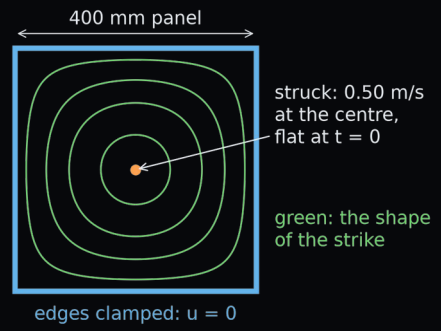
Exercise 7.2

OPTIONAL · THREE NOTEBOOKS, ONE PER TYPE

- **1** — heat: the silicon die after a power pulse
- **2** — Helmholtz: the stator slot and its runaway
- **3** — wave: the struck panel
- **each** — the exact solution, finite differences, a PINN
- **each** — a report and a mini project, MP7.2A – C







THE QUESTION EACH NOTEBOOK ANSWERS

1	does the PINN meet 0.02 K?
2	at what current does it run away?
3	why is the wave the hardest?

REPORT QUESTION, TO CONCLUDE

Finite differences met the agreed accuracy on all three, in milliseconds. Which type was hardest for the network, and what does the type of an equation tell you before you train?

SUMMARY

Key Takeaways

STEADY	
what it is	elliptic: every wall felt at once
function	Poisson a source; Helmholtz a mode
train, tune	a unit norm, and a floor on λ
vs in time	no initial condition, no time input
used for	conduction, potentials, modes

IN TIME	
what it is	heat parabolic, wave hyperbolic
function	diffusion decays; a wave travels
train, tune	the initial field; sine for waves
vs steady	time an input; one or two ICs
used for	a die after a pulse, a panel

Next — L8.1: steady heat on real geometry, a plate with a cooling hole.

Questions

1 – 2 · STEADY EQUATIONS

1 **How does the type of the equation decide what the network must be given?**

Poisson's equation has no time: the position only, and a condition on every wall. The heat equation adds time and one initial field; the wave equation needs two, the field and the velocity. With one missing, the loss still reaches zero, but at a wrong field. *Go deeper: the panel without its initial velocity (Ex_07.2 notebooks 1 – 3); Liu ch. 2.*

2 **Why does a network trained on the Helmholtz equation return a zero field, and how is that prevented?**

Zero satisfies the equation and the walls with a residual of exactly zero, so gradient descent finds it first. A normalisation term fixes the size of the mode, and a floor keeps the eigenvalue away from zero. Each further mode is asked to be orthogonal to those found. *Go deeper: the Helmholtz supplement of Ex_07.2; Liu ch. 6.2, 6.4 – 6.5.*

3 – 4 · EQUATIONS IN TIME

3 **How does a network represent a field in time, and why is an early error worse for a wave than for heat?**

Time is one more input, and the network returns every position and instant at once. Heat damps every feature, so an early error fades. A wave keeps its shape, so an early error travels with it: the errors fall with time on the die and grow on the panel. *Go deeper: the error against time (Ex_07.2 notebooks 1, 3); Liu ch. 4.1, 5.1.*

4 **How are the initial conditions given to a network in time, and why is the wave residual scaled?**

A condition with a formula is built into the trial function. The initial velocity cannot be, so it is a loss term. The panel's acceleration is hundreds of times its displacement, so the residual is divided by its scale; otherwise the network settles on no motion at all. *Go deeper: train without the scaling (Ex_07.2 notebook 3); Liu ch. 4.2, 5.2.*

NEXT

L8.1 — Stationary Heat

Poisson's equation with real units, a convection wall and a plate with a cooling hole.

BEFORE THE NEXT SESSION

read	Liu ch. 4.1, 5.1 – 5.2, 6.1
run	Ex_07.2, all three notebooks
bring	the three errors and times
think about	which type your own problem is

Block L7 is finished: the method, and the four equations it is applied to.